

GENERAL SPECIFICATIONS G6.4

Instrument and Control Panel Construction

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G6.4 INSTRUMENT AND CONTROL PANEL CONSTRUCTION

G6.4.1 General

- (a) Unless otherwise specified, all instrument panels, instrument enclosures, instrument cubicles, remote i/o panels, field network interface panels, PLC panels, electrical interface cabinets, control panels, consoles, and the like shall be free-standing floor mounted units, all referred to herein as panels.
- (b) If the Contractor proposes panels other than free-standing floor mounted types then a detailed submission complete with substantiation shall be given to the S.O. for approval on a case by case basis.
- (c) The panels shall be designated for either indoor or outdoor locations with enclosure requirements as set out below.
- (d) PLC Remote i/o panels not located in XLV rooms or switchrooms, shall be fitted with limit switches connected to a PLC digital input to provide an alarm on SCADA whenever a panel door is opened. For panels with multiple doors, limit switches may be series connected to a common PLC input.
- (e) Various types of instrument and control panels are envisaged as follows:
 - (i) XLV Room Rack Enclosures;
 - (ii) Rack Enclosures for distributed Networking Equipment;
 - (iii) Indoor Panels for air-conditioned rooms;
 - (iv) Indoor panels not for plant areas; and
 - (v) Outdoor panels.

G6.4.2 XLV Room Rack Enclosures for PLC's, Servers, and Networking Equipment

- (a) Panels for PLCs, servers and networking equipment, and for fibre-optic cable patch panels (FOBOTs), which are located in air-conditioned XLV rooms or server rooms or network administration rooms, shall be 19inch racks, 42RU tall in accordance with IEC 60297, with mesh/honeycomb for sides and doors to facilitate ventilation.
- (b) Doors shall be colour coded to distinguish between process and non-process systems:
 - (i) Process Systems : Light Grey(RAL 7035)
 - (ii) Non-Process Systems (excluding fire systems) : Black(RAL9011)
 - (iii) Fire Systems : Red (RAL3000)
- (c) Servers shall be mounted on draw-out rails, with a folding cable management device for each server.
- (d) Rack enclosures shall have two power feeds.
- (e) Panels for remote i/o in XLV rooms may follow the requirements for outdoor panels if the Contractor desires to adopt a standard design for remote i/o panels.

G6.4.3 Rack Enclosures for Distributed Networking Equipment

- (a) Panels for selected networking equipment may be distributed throughout plant areas as per the Particular Specification. These panels shall be in sheltered indoor plant locations, but not necessarily in air-conditioned rooms.
- (b) Rack enclosures for distributed networking equipment shall be wall-mounted 316SS IP66 rated 19inch racks, in accordance with IEC 60297, of 11 or 15RU size as appropriate.

G6.4.4 Instrument and Control Panel Construction (Indoors)

- (a) Enclosures for indoor locations shall be manufactured from cold rolled and annealed zinc-coated mild steel sheet, suitably braced and stiffened as necessary with flat bar or angle to form a rigid structure.
- (b) The degree of protection shall be suitable for the location but as a minimum shall be:
 - (i) IP 31 in switchrooms, XLV rooms, and other air-conditioned rooms; and
 - (ii) IP 54 for other indoor locations.
- (c) Panels for remote i/o in XLV rooms may follow the requirements for outdoor panels if the Contractor desires to adopt a standard design for remote i/o panels.
- (d) Where ventilation fans/openings are included in the panels, louvers and dust filters shall be provided to ensure the specified minimum degree of protection is maintained.
- (e) Panel fronts shall be flat and free from bow or ripple. Exterior corners and edges shall be rounded, or welded and ground, to give a smooth overall appearance. Flanged edges shall be straight and smooth.
- (f) The thickness of sheet metal shall be chosen with due regard to the enclosure size, number of cut-outs, instrument weight, centre of gravity, and method of fabrication, subject to the following minimum values:

Instrument/equipment bearing surfaces	3 mm
Gland plates	3 mm
Pneumatic distribution plates	3 mm
Internal mounting plates	3 mm
Doors, covers, filler panels	2 mm
- (g) The method of construction shall not involve the use of externally-visible fixing bolts and screws. There shall be no dust or water-collecting crevices.
- (h) Stiffeners and supporting framework inside panels shall be provided where necessary. Framework shall be hinged or fixed, suitable for the installation of instruments, components and internal equipment for which it is provided and located to give easy access to adjacent equipment.
- (i) If the enclosure is constructed in sections, they shall be designed for ease of assembly and necessary nuts, bolts and washers shall be supplied in the same shipment.

- (j) Unless specified otherwise, each enclosure shall be mounted on a self-draining base frame fabricated from 150 mm deep, hot dip galvanised steel channel section which shall be drilled or provided with clamps for bolting to the floor. The base frame shall be set back from the panel front face to give a toe space of not less than 25 mm. The outside of the base frame shall be covered with an approved kicking strip.
- (k) Where wall or surface-mounting is specified, the fixings shall be external to the enclosure which shall be mounted not less than 10 mm away from this surface.
- (l) Unless otherwise specified, all cable entry shall be from the bottom.
- (m) For "walk-in" type panels a chequer-plate floor shall be provided inside and above the level of the base frame with cable access. A raised cable termination chamber with horizontal removable gland plates and vertical removable access panels shall be provided for field cabling.
- (n) Panels containing pneumatic or other instruments using a fluid as the transmission medium shall have distribution plates with bulkhead unions for the termination of internal and external pipework.
- (o) No equipment other than front-of-panel items shall be mounted on panel surfaces.
- (p) If electrical and non-electrical instruments are mounted in the same panel, the panel shall be sub-divided internally to separate the electrical and non-electrical sections. All connections shall be arranged to ensure that no accidental damage to cabling or electrical components can occur in the event of failure of any non-electrical component or connection.
- (q) Provision shall be made for safe and easy handling during transit and installation. If lifting eyes are provided they shall be reversible and panel tops shall be reinforced where necessary.
- (r) Where required for equipment to be installed at a future date, space shall be allocated and cut-outs with removable masking plates, brackets, supports, wiring, piping etc. as necessary, shall be provided.

G6.4.5 Instrument and Control Panel Construction (Outdoors)

- (a) Analysers and other instrumentation & control equipment, including PLC remote i/o, that has been designated for outdoor locations shall be installed in a 316L stainless-steel panel.
- (b) The panel shall be rigid and secure and shall have a minimum degree of protection of IP 65. Panels shall be mounted on a concrete base, either directly or via a stand, with provision for cable access.
- (c) Where ventilation fans/openings are included in the panels, louvers and dust filters shall be provided to ensure the ingress protection rating specified is maintained. Panel design shall take into consideration and protect against windblown rain.
- (d) The roof section of the enclosures shall slope downwards away from the door side. Large plane surfaces shall have adequate reinforcing to ensure rigidity.

- (e) A concrete or concrete slab hard-standing area of minimum width 600mm shall be provided in front of the panel.
- (f) A weather canopy constructed from hot dipped galvanised steel cladding shall be provided above the panel. The canopy shall have a suitable overhang to allow sheltered operation of any door mounted controls and to prevent direct sun striking the panel.
- (g) Door locks, handles and hinges shall be of a high tensile strength, non- corroding alloy with stainless steel pins and through fixing bolts. Doors shall be fitted with rigid self-locking stays arranged to hold the door open at a minimum of 100 degrees and door stay release.
- (h) Small housings less than 0.5m³ volume for instrumentation equipment shall be of stainless steel construction. Hinges, handles and locks shall apply as for large housings.
- (i) Ground mounted small housings shall be installed on a concrete base as for large housings. Wall mounted housings shall be provided with external fixings and shall be a minimum of 10mm from the mounting surface.
- (j) Threaded studs shall be incorporated into the structure of the housing for the mounting of sub-frames or backing plates or boards on which all internal equipment shall be installed. Fixing bolts through the skin of the enclosure shall not be accepted.
- (k) The front of externally - visible instruments and any window shall be of toughened glass and shall not reduce the IP degree of protection of the housing. Where practical, outdoor panels shall be compartmentalised with separate doors such that cable termination can be carried out whilst instruments and equipment shall remain protected.
- (l) PLC hardware shall be protected from dust and static electricity as well as from Hydrogen Sulphide (H₂S).

G6.4.6 Panel and Rack Enclosure Arrangement

- (a) The height of panels above finished-floor level shall not exceed the following limits:

Control consoles and desks	1400 mm
Other panels	2130 mm
- (b) Enclosures shall be symmetrically arranged so far as possible with projections kept to a minimum.
- (c) Front of panel equipment shall be mounted so that their centres fall within the following limits of height from the floor:

	<u>Minimum</u>	<u>Maximum</u>
Indicators	900 mm	1800 mm
HMI units	900 mm	1400 mm
Signal lamps	750 mm	2000 mm
Manual Controls	750 mm	1500 mm

- (d) Controls, switches and push buttons shall be positioned below or adjacent to any associated reading instrument that they influence.
- (e) Each panel which contains a PLC CPU shall have a fold down shelf, to allow a programming device to be used comfortably.

G6.4.7 Panel and Rack Enclosure Access

- (a) The arrangement of equipment within panels shall permit easy access for installation and maintenance.
- (b) Panels shall have hinged access doors fitted with recessed lockable handles. Hinges of lift-off doors shall be designed so that one shank engages before the other for ease of fitting. Wherever necessary, removable access covers secured by quick release fasteners shall be provided to ensure ease of access to all installed apparatus.
- (c) Locks shall have Euro-profile single cylinder key barrels. Temporary key barrels shall be provided during construction and commissioning and these shall be replaced with the final keying system prior to practical completion.
- (d) The final keying system details shall be agreed with the S.O. and shall be Dormakaba pextra, Medeco m3, BiLock or equivalent.
- (e) All doors of each category of panel shall use the same key. Panels categories, for keying purposes shall be:
 - (i) PLC CPU Panels, Process Server Racks and Communication Racks;
 - (ii) Non-Process Network Racks and Non-Process Network Panels including Distributed Network Equipment Racks;
 - (iii) PLC RIO Panels (excluding IWMF);
 - (iv) Instrument and Analyser Panels; and
 - (v) RIO Panel at IWMF.
- (f) Marshalling Panels and Junction Boxes do not require security key locks and shall have standard “double-bit” panel locks.
- (g) Minimum clearances from walls and other panels shall be as per the Particular Specification.

G6.4.8 Panel and Rack Enclosure Finish

- (a) Panels shall be subjected to a comprehensive system of preparation and protective coating. The finish coat of paint shall be applied by the electrostatic process and shall be to the colours specified or as approved by the S.O..
- (b) The preparation and painting system shall be suitable for the environment in which the panels are to be installed.
- (c) Full details of the preparation and painting system for panels not located within air-conditioned rooms shall be submitted to the S.O. for approval.
- (d) Stainless Steel panels need not be coated.

G6.4.9 Panel and Rack Enclosure Safety

- (a) ICA equipment shall be designed and constructed to provide a maximum standard of safety for operational and maintenance personnel. Extra low voltage shall be used wherever practical. Low voltage (230 Vac typically) shall generally be limited to the supply to 24 Vdc power supply units and to PLC chassis power supply units and for servers and backbone switches. All other panel devices shall be powered from 24 Vdc or 48 Vdc for power over Ethernet (PoE) devices.
- (b) Within ICA panels, low voltage terminals shall be grouped together and the area covered by an additional shroud to prevent accidental contact. Warning labels shall be fitted to the shroud. Shrouds shall have a minimum degree of protection of IP 30, this is in addition to the inherent IP protection of terminals. Low voltage terminals and wiring shall not be interspersed with extra low voltage terminals and wiring. Separate looms and trunking shall be used.
- (c) Covers to compartments incorporating exposed live conductors exceeding extra low voltage shall be bolted and shall be fitted with a warning label.

G6.4.10 Panel Earthing

- (a) A continuous copper earth bar of not less than 25 mm x 6 mm cross section shall run the full length of each panel and shall be securely fixed and bonded electrically to the main frame of the panel to function as a protective earth. The cable gland plates and the earth bar shall be provided with suitable brass terminals of not less than 6 mm diameter for connecting the metal cladding or armouring of all incoming and outgoing cables to the plant electrical earthing system.
- (b) A second continuous copper earth bar of not less than 25 mm x 6 mm cross section, electrically isolated from the steelwork of the panel and metal cladding and armouring of cables, shall be provided as an instrument earth. This shall be used for earthing any signal earth connections on instrumentation and control devices and for earthing the screen(s) of each instrument cable. The instrument earth bar shall have sufficient brass terminals for each instrumentation and control device and for the screen of every shielded cable plus 25% spare terminals.
- (c) In multi-section panels, each protective earth bar shall and each instrument earth bar shall be electrically bonded to their corresponding bars in the adjacent section(s).
- (d) A flexible protective earth connection shall be provided to all enclosure doors. Earthing via the door hinge shall not be acceptable.
- (e) Panel instrument earth bars shall be connected by 10mm² cable to the main instrument earth bar in each facility.
- (f) Protective earths shall be connected in accordance with Singapore Standard CP5.
- (g) Rack enclosures for distributed networking equipment shall require only protective earthing in accordance with CP5 and shall not require separate instrument earth arrangements.

G6.4.11 Panel and Rack Enclosure Isolation

- (a) Clearly-labelled switch-disconnectors shall be provided for each incoming power supply. For ICA panels these shall generally be located inside the panel.
- (b) Openable isolating links of an approved type shall be provided in any i/o circuit that could still be alive when the panel main switch-disconnectors are in the "off" position. Such links or devices shall be properly shrouded and labelled.

G6.4.12 Circuit Protection

- (a) Adequate facilities for isolation and protection by miniature circuit-breaker or fuse for each instrumentation and control circuit and sub-circuit (including i/o circuits/loops) shall be provided and shall be so arranged that any interruption causes minimum disruption to plant operation, operates the appropriate alarm and cannot result in any unsafe operating condition.
- (b) All miniature circuit-breakers and fuses shall comply with Electrical Section of General Specification. Single pole circuit isolation shall not be allowed.
- (c) Bases for solid links shall not be interchangeable with those for fuses. Fuses and links in the same circuit shall be mounted opposite each other in separate adjacent rows and shall not alternate in the same row. At least 10% unallocated miniature circuit breakers or fuses and links shall be provided in each panel distribution board. Miniature circuit breakers and fuses of similar size and rating shall be of the same make and type.
- (d) At least 10% spare fuses and links of each rating shall be provided and fitted in clips inside the panel.
- (e) Each instrument requiring a power supply shall be individually wired and protected so that in the event of a failure in one circuit the remainder are unaffected. Power supply circuits shall be of sufficient rating that any protective device may operate without reducing the voltage at the terminals of any other component to an unacceptable level. Remote alarms shall be operated on failure of the electrical supply to a panel. There shall be protection discrimination between protective devices and their corresponding upstream device.

G6.4.13 Panel and Rack Enclosure Internal Wiring

- (a) Panel circuits shall be segregated into the following categories:
 - (i) Group 1 containing:
 - 1) Circuits exceeding extra-low voltage (i.e. supplied at 230 Vac or 110 V dc); and
 - 2) AC current signals above 50 mA (e.g. current transformer circuits).
 - (ii) Group 2 containing:
 - 1) Signals from conventional electronic transmitters and controllers (e.g. 4-20 mA);
 - 2) Circuits to solid state devices (excluding those in Group 3);
 - 3) Digital i/o and control signals;

- 4) DC power supplies (ELV); and
- 5) Fieldbus networks.
- (iii) Group 3 containing:
 - 1) Signals from thermocouples;
 - 2) Signals from resistance thermometers and retransmitting slide-wires;
 - 3) Analogue signals from analytical sensing elements and strain gauges; and
 - 4) For Group 3 wiring, internal connections to the instruments shall be made by one of the following methods;
 - either: The twisted, screened conductors of the external cable shall be led direct to their appropriate instruments via ducting systems installed for this purpose during construction of the panel.
 - or: The conductors of the external cables shall be terminated on terminals segregated from all other categories and the connections to the appropriate instruments shall be made using twisted pairs with individual screening installed for this purpose during construction of the panel.
- (b) Internal wiring for all circuits in Group 2 except those sharing a common connection shall be multi-stranded, twisted pair, 0.5 mm² minimum copper conductor with HDPE or PVC insulated cable of adequate grade and rating in accordance with BS 6004.
- (c) Wiring for circuits in other Groups or sharing a common connection shall be run in stranded, 1.0 mm² minimum copper conductor with 600 V grade, PVC insulated cable of adequate grade and rating.
- (d) Wiring sheath colours shall be black for a.c. circuits, and grey for d.c. circuits (excluding thermocouple circuits) and light blue for intrinsically safe circuits.
- (e) Separate ducts, trunking, cable looms, tray work etc. shall be provided within the panel for each category with at least 150 mm between parallel paths of Group 1 and those of any other Group.
- (f) The hazardous area (field) side of intrinsically safe circuits and their terminals shall be segregated from all other circuits and terminals, but shall otherwise be generally treated as for Group 2 circuits. Trunking for intrinsically safe wiring shall be blue in colour.
- (g) All wiring shall be neatly and securely fixed by insulated cleats, bunched and secured by approved plastic strapping or run in approved insulated wiring trunking or corrosion proof flexible tubing. Not more than 75% of the capacity of trunking, ducts, looming, or tubing shall be used. Insulated earth wiring shall be so arranged that access to any equipment or connection point or the removal of any item of equipment is unimpeded. Wiring for future equipment shall be secured and terminated on terminal blocks. Lacing for wiring looms shall be of rot-proof cord or plastic strips.

G6.4.14 Panel and Rack Enclosure Wiring Identification and Termination

- (a) Identification ferrules shall be fitted at both ends of each wire. The numbers or letters used shall correspond with the appropriate wiring diagram. The ferrules shall be of plastic insulating material with permanent black characters on a gloss white or yellow background unaffected by oil or water. They shall be so arranged that they can be read logically from left to right when viewed normally.
- (b) The system of wire identification shall be such that wires in the same circuit on opposite sides of a terminal shall have the same reference, and this system shall be continued through all external cabling.
- (c) Terminal ferrules (spade or tongue, crimped connections) shall be provided on each conductor.

G6.4.15 Panel and Rack Enclosure Terminal Blocks

- (a) A terminal block shall be provided, as the interface between the corresponding conductors of each internal and external wire and each internal and external connection except for:
 - (i) Power supply cable to switch-disconnector;
 - (ii) Group 3 wiring when connected directly to the signal processing device; and
 - (iii) Intrinsically safe circuits which terminate directly to the IS protection barrier.
- (b) Terminal arrangement
 - (i) The terminal blocks shall be mounted vertically where possible and not nearer than 250 mm to the floor or less than 250 mm from an incoming cable gland. Terminal block rows shall be spaced apart by not less than 150 mm and arranged to permit convenient access to wires and terminals and to enable ferrule numbers to be read without difficulty.
 - (ii) Circuits shall be grouped on the terminal blocks according to the classification given in section G6.4.13 which shall be clearly marked along the corresponding section of each terminal board. Groups of different voltages on the same board shall be separated by insulated barriers. If terminals are used for intrinsically safe circuits, they shall be blue in colour.
 - (iii) The number of terminals fitted shall be sufficient to allow all installed PLC I/O points to be wired down to terminals. Sufficient spare terminal rail shall be installed to allow subsequent installation of terminals and wiring down of the required 30% (minimum 4) spare I/O slots.
 - (iv) All connections shall be made from the front of terminal blocks and no live metal shall be exposed at the back.
 - (v) Terminal blocks for input and output signals shall be of the type which permit the connection of a test milli-ammeter or continuity meter without disconnecting any wiring and shall have an integral means of isolating the two parts of the circuit. Terminal blocks for power supplies for equipment external to the panel shall permit the isolation of the item of external equipment without affecting the operation of any other circuit within or outside the panel.

- (vi) No more than one core of external cables, or one internal wire shall be connected to any terminal. If terminal blocks are used as common points for two or more circuits, individual terminals with the appropriate number of permanent cross-connections shall be provided. The lengths of exposed cable cores shall be sufficient to reach any terminal in the appropriate row or rows. The cores shall be formed into a neat loom and a separate loom shall be provided for each cable. Identification ferrules as specified shall be fitted on each core of all external cables and on each internal wire.
 - (vii) The size of the terminals shall be appropriate to the size and current rating of the cable cores which are to be connected to them.
 - (viii) All the terminal boards shall be provided with covers of transparent insulating material that does not sustain combustion and shall be sectionalised where possible to give access to groups of terminals without uncovering all boards.
- (c) Terminals blocks
- (i) All terminal blocks shall be screwless spring cage clamp type for copper conductors from 0.08mm² up to 35mm². The terminal blocks shall conform to IEC 60947-7-1. Individual terminal block shall have conductor entry holes that are closed on all sides, to provide optimum mechanical support to the cable in all directions and allowing insertion of a fine stranded cable without special preparation.
 - (ii) Multiple deck terminals shall not generally be used. Any use shall be approved by S.O.
 - (iii) In all cases, unless stated otherwise, the terminal blocks shall be of the knife disconnect type with 2.3mm test sockets at each clamping unit and with clear marking facilities on the movable knife disconnect or of the disconnection tab type with coloured tab to indicate switching condition for the testing and isolation purpose of circuits.
 - (iv) Where protection is required, fuse terminal blocks shall be integrated with LED for indication of blown fuse. The fuse holder shall be pluggable type with pull-tab, touch proof test slots and with marking facilities.
 - (v) Only one wire shall be connected to each clamping unit of a terminal block. Where additional connections are required, additional terminal blocks shall be provided. Multi-conductor terminal block (3-conductor or 4-conductor) with additional clamping units or touch proof push-in type jumpers may be used where there is space constraint and upon approval on a case by case basis by the S.O..
 - (vi) Earth terminals as well as end stops shall be of the screwless, snap-on type. When mounted, earth terminals shall make an automatic connection to the rail by means of a grounding foot with an integrated spring to ensure a secure ground connection and prevent lateral displacement on the DIN rail.
 - (vii) Samples of terminal blocks to be submitted to S.O. for approval.

- (d) Materials
 - (i) Terminal mouldings shall be to BS 1322. Terminal rails shall be hot-dip galvanised. Current bars between the two connection points of each terminal block shall be of copper or brass with tin/lead alloy plating. All steel parts shall be zinc-plated and passivated with a yellow chromate layer.
- (e) Spare Terminals
 - (i) Each row of terminal blocks shall contain at least 20% spare terminals over the number required for terminating all cores of external cables in that row.
- (f) Numbering
 - (i) Terminal blocks shall be numbered consecutively in a sequence different from that used for identifying wiring. The terminal numbers, voltage grouping and terminal board layout shall correspond precisely with wiring diagrams so that quick and accurate identification of wiring can be made.
 - (ii) Terminal groups must be adequately identified and shall correspond precisely with wiring diagrams.

G6.4.16 Panel Heating

- (a) Anti-condensation heaters are not required unless stated in Particular Specifications P6 ICA, or shown in Drawings.
- (b) Heaters shall be mounted as low as possible and shall be located to ensure free air flow over the heating element and to have no detrimental effect on temperature sensitive devices or adjacent cabling.
- (c) Heaters shall be fitted with a safety guard the surface temperature of which shall not exceed 60 °C.
- (d) Heaters shall be rated to ensure condensation does not occur under the ambient air temperature and relative humidity conditions prevailing at the location where the panels are to be installed.
- (e) Heaters shall be complete with protective device, ON/OFF switch and thermostat.

G6.4.17 Panel Lighting & Socket Outlets

- (a) Each floor-mounted panel shall be adequately illuminated internally, as evenly and as free from dazzle as possible, by fixed 24 Vdc LED lighting controlled from totally enclosed light switches and by totally enclosed door-operated switches positioned so as not to interfere with access.
- (b) There shall also be one installed LED inspection lamp per three metres of panel length or part thereof with adequate flexible connection cable to reach any point in the panel. The control switch for an inspection lamp shall form part of the lamp assembly.
- (c) Lighting circuit protective devices shall be independent of any instrumentation and control circuit and designed to allow lamps to be replaced safely. Circuit breakers for lighting shall have ratings providing discrimination with upstream circuit-breakers.

- (d) Cabinets having a 230 Vac supply shall be fitted, internally, with standard twin 13A switched socket outlets at the rate of one socket for each two metres of panel length with a minimum of two sockets. These shall be fitted with a residual current device (RCD). Circuit breakers for these outlets shall have ratings providing discrimination with upstream circuit-breakers. Ratings are not required to exceed 6A.
- (e) Socket outlet circuit protective devices shall be independent of any instrumentation and control circuit and shall be sized with a view to the limited capacity of any upstream UPS. It shall not be possible to trip the supply to the panel merely by connecting a heavy load to an internal panel socket outlet.

G6.4.18 Panel Ventilation

- (a) Each panel shall have sufficient fans to ensure adequate ventilation of the equipment.
- (b) Fans shall be controlled by a suitably labelled enclosed switch mounted internally in an accessible position. Circuit breakers for fans shall have ratings providing discrimination with upstream circuit-breakers.
- (c) Fans shall be mounted with their axis horizontal and shall be arranged to draw clean air into the panel. Air entries shall have filters which can be renewed from outside the panel and shall be designed to prevent entry of rain, spray, injurious fluids, sand or dust.

G6.4.19 Panel Piping and Tubing

- (a) Panels containing equipment using a supply of instrument air shall have a common air pressure-regulating station with duplicate pressure reducing valves and filters. The pressure reducing station shall also include isolating valves upstream and downstream of each filter/regulating valve set, pressure indicator for the low-pressure header and a pressure indicator for the high-pressure pipework. The pressure-reducing station components shall be mounted in a clear space inside the panel, supported on a suitable framework between the lower horizontal row of instruments and the low-pressure header.
- (b) All piping, fittings and valves downstream of the pressure reducing station shall be of 316 stainless steel. PTFE tape shall not be used downstream of the main filters.
- (c) The low-pressure header shall be 316 stainless steel and shall be near the panel floor with drain valves and tundishes piped to a drain. Branch air headers shall be of 316 stainless steel (15 mm dia. minimum) and shall run vertically from the header to the instrument. The low-pressure header and each branch shall have a 6 mm minimum, 316 stainless steel shut-off valve for each instrument requiring an air supply and a compression coupling for each air purge connection. At least 10% spare connections for possible future instruments shall be provided in each panel section. Any header dismantled before shipment shall have 316 stainless steel unions or flanges at each panel section junction.

- (d) Panel-mounted instruments shall be piped to bulkhead fittings on a gland plate during assembly at the manufacturer's works. Piping shall be segregated from wiring so that any leakage is harmless. Each panel-mounted pressure gauge shall have a stainless-steel flush-mounted shut-off and fine-regulating valve mounted vertically below. A drip tray shall be provided below each row of gauges.

G6.4.20 Panel and Rack Enclosure Labels

- (a) Labels shall be provided for every panel to describe the duty or otherwise identify the panel and its sections and every instrument, component and item of equipment mounted internally and externally. Additionally, labels shall be provided to list the calibrated range of each instrument.
- (b) Wording shall be clear, concise and unambiguous and shall be subject to the S.O.'s approval before manufacture. Each label shall be permanently secured to the surface near the item to which it refers.
- (c) Labels shall be manufactured from rear engraved plastic.
- (d) Rear engraved plastic labels shall be clear with black engraving unless otherwise specified. Internal labels shall be inscribed with the component description given on the associated circuit diagram.
- (e) Labels for fuses shall have the fuse link rating inscribed.
- (f) Internal labels shall not be fixed to components (unless there is also a corresponding label fixed to the gearplate) or cable trunking covers.
- (g) Danger and warning labels shall comply with BS 5499, and shall unless otherwise approved by the S.O., be manufactured from rigid plastic.
- (h) Labels shall lay flat to the surface and shall be secured by non-rusting screws.
- (i) Labels shall be in English unless otherwise specified.
- (j) Embossed plastic and paper labels shall not be accepted.
- (k) Labels conforming to the requirements of the preceding paragraphs or other approved means shall be provided to:
 - (i) Describe or identify circuits or circuit components;
 - (ii) Identify dc. Polarity;
 - (iii) Warn of hazards; and
 - (iv) Wherever elsewhere specified.
- (l) Unless otherwise specified, all engraving shall be in plain block letters, 4 mm high. The minimum practicable number of different sizes shall be used.

G6.4.21 Push-Buttons and Indicator Lights

- (a) Push-buttons and indicator lights shall comply with the relevant section in the electrical specification.
- (b) Push-buttons in control circuits shall have shrouds, guards or other suitable means for preventing inadvertent operation.
- (c) Pushbuttons for lamp test and other necessary purposes shall be provided.
- (d) If legends are engraved on the pushbuttons they shall be clear and concise and shall be approved by the S.O..

G6.4.22 Panel and Rack Enclosure Internal Power Supplies

- (a) Where required, panels shall have integral power supply units to produce smooth, regulated, 24 Vdc from the panel 240 Vac input.
- (b) For PLC panels, Remote i/o panels, and communication racks (other than FOBOT racks having no active components), power supply units shall be provided in a 100% + 100 % duty-duty arrangement with a diode sharing module. There shall be a monitoring contact connected to PLC i/o to indicate the failure of either power supply unit.
- (c) Each power supply unit shall be rated with at least 30% spare capacity.
- (d) Power supply output current capability shall be sufficient to ensure reliable tripping of final sub-circuit protection devices.

G6.4.23 Factory Testing

- (a) Panels and rack enclosures, and their internal equipment, shall undergo a factory acceptance test (FAT) in Singapore prior to delivery to site
- (b) The Contractor shall provide the S.O. with two weeks-notice of any panel FAT and shall allow the S.O. to witness the FAT.
- (c) Proposed test plans and documentation shall be submitted to the S.O. for approval two-weeks prior to the FAT.
- (d) Testing of remote i/o panels shall include use of PLC CPU emulator to allow PLC outputs to be driven and PLC inputs to be read. The emulator shall be provided by the Contractor.

END OF SECTION